



OASIS INFOBYTE

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SIP PROGRAM

JAVA DEVELOPMENT INTERNSHIP PROJECT

Project: Number Guessing Game

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Problem Statement

The Number Guessing Game is a console-based Java application in which the computer generates a random number and the user attempts to guess it. After each guess, the application provides a hint indicating whether the guess is too high or too low, while tracking the available attempts.

Objectives

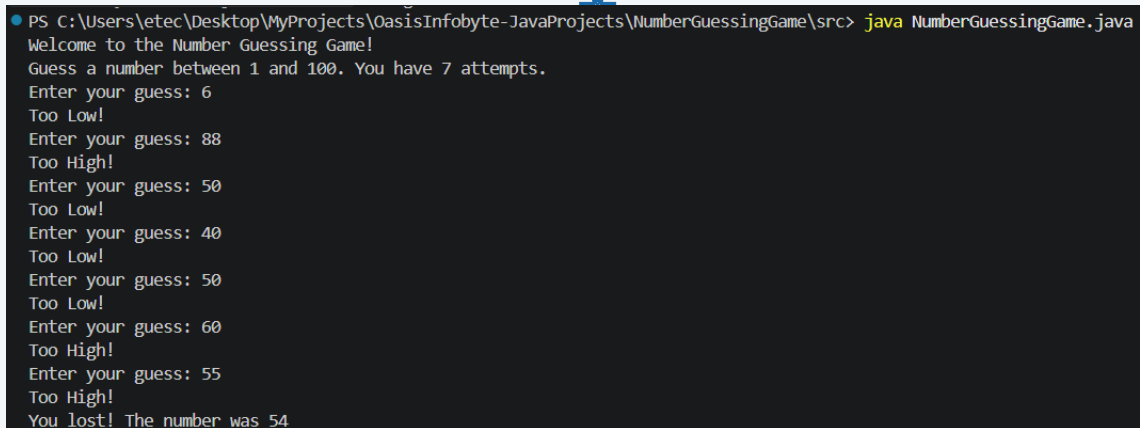
- Generate a random number within a defined range.
- Accept and validate user input.
- Provide clear high/low feedback after each guess.
- Track the number of attempts and determine win/loss conditions.

Tech Stack

- Java console application.
- `java.util.Random` for random number generation.
- `java.util.Scanner` for keyboard input.
- Loops and conditional statements for game flow and decision-making.

Implementation

The application generates a secret number between 1 and 100 and repeatedly prompts the user for a guess. The program compares each guess with the generated number and displays “Too Low!” or “Too High!” when the guess is incorrect. The game ends when the user guesses correctly or exhausts the allowed attempts.



```
PS C:\Users\etec\Desktop\MyProjects\OasisInfobyte-JavaProjects\NumberGuessingGame\src> java NumberGuessingGame.java
Welcome to the Number Guessing Game!
Guess a number between 1 and 100. You have 7 attempts.
Enter your guess: 6
Too Low!
Enter your guess: 88
Too High!
Enter your guess: 50
Too Low!
Enter your guess: 40
Too Low!
Enter your guess: 50
Too Low!
Enter your guess: 60
Too High!
Enter your guess: 55
Too High!
You lost! The number was 54
```

Figure 1. Number Guessing Game – console gameplay output

Conclusion

The project demonstrates fundamental Java programming concepts including variables, input handling, random number generation, loops, conditionals, and basic game logic. Future enhancements may include a graphical user interface, multiple difficulty levels, score tracking, and replay functionality.